

Module 7

Hints and Solutions

Section A – Fundamental Concepts

Problem 2

(a) $\Re(s) > -2$

(b) $\Re(s) < 3$

(c) $-2 < \Re(s) < 2$

Problem 3

(a) $\Re(s) > -3$, Exists

(b) $\Re(s) > 2$, Exists (only in ROC)

(c) $\Re(s) > 0$, Exists

(d) $\Re(s) > 0$, Exists

Problem 4

(a) $\Re(s) > -2$

(b) $\Re(s) > -4$

(c) $\Re(s) > 0$

(d) $\Re(s) > 0$

Problem 5

ROC: $\Re(s) > 0$

```
import numpy as np
import matplotlib.pyplot as plt

# Time axis
t = np.linspace(-1, 3, 1000)

# Rectangular pulse:  $x(t) = 1$  for  $0 \leq t < 2$ , otherwise 0
x = np.where((t >= 0) & (t < 2), 1, 0)

# Plot
plt.figure(figsize=(8,4))
```

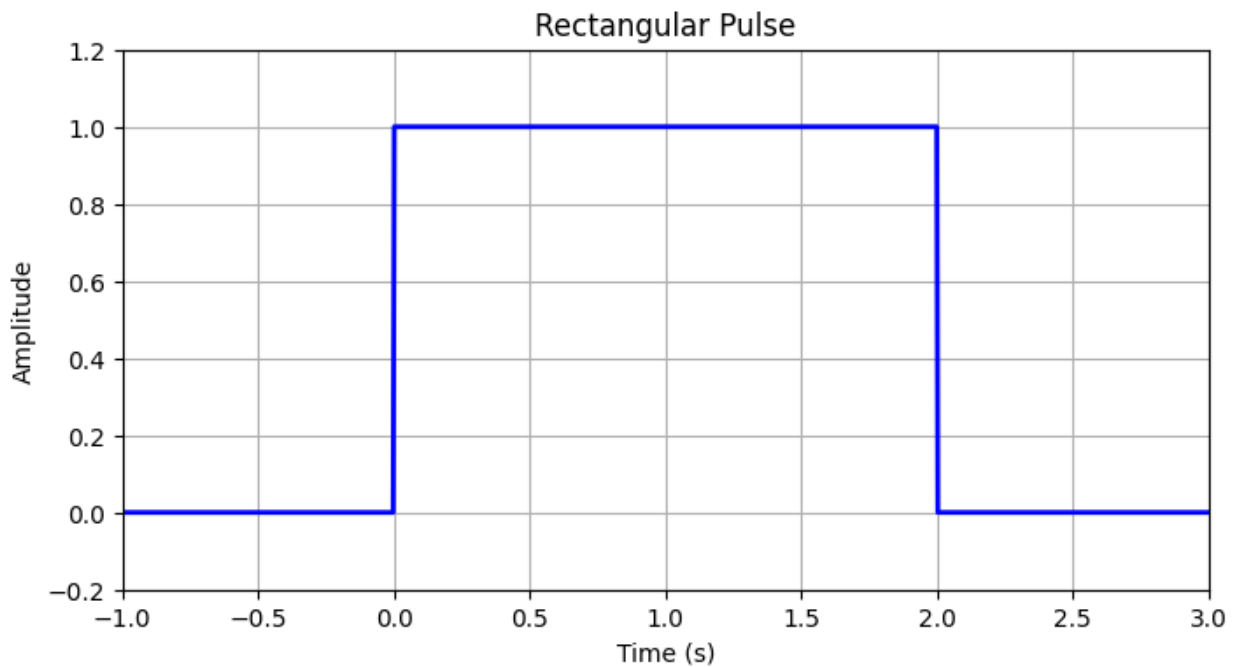
```
plt.plot(t, x, color='blue', linewidth=2)

plt.title("Rectangular Pulse")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")

plt.xlim(-1, 3)
plt.ylim(-0.2, 1.2)

plt.grid(True)

plt.show()
```



Section B – Properties of Laplace Transform

Problem 6

$$X(s) = \frac{3}{(s+1)} + \frac{2}{(s+3)}$$

```
import numpy as np
import matplotlib.pyplot as plt

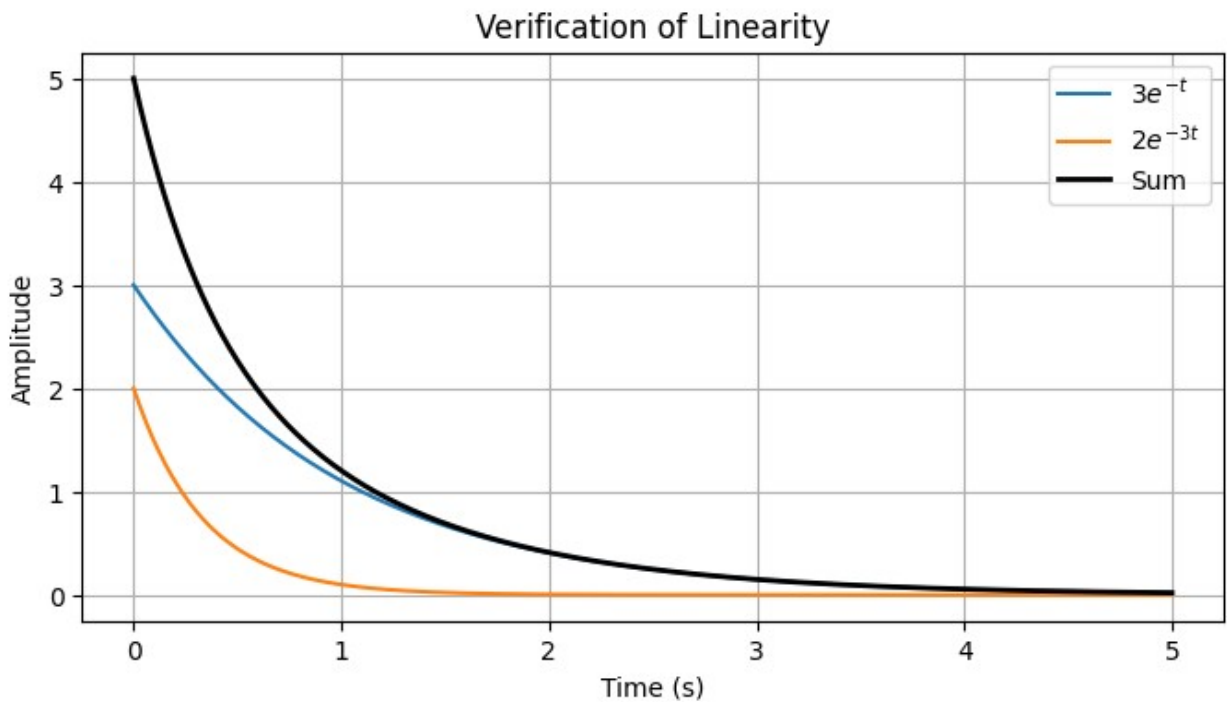
t = np.linspace(0, 5, 500)

x1 = 3*np.exp(-t)
x2 = 2*np.exp(-3*t)

x = x1 + x2
```

```
plt.figure(figsize=(8,4))
plt.plot(t,x1,label=r'$3e^{-t}$')
plt.plot(t,x2,label=r'$2e^{-3t}$')
plt.plot(t,x,'k',linewidth=2,label='Sum')

plt.title("Verification of Linearity")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.legend()
plt.grid(True)
plt.show()
```



Problem 7

```
import numpy as np
import matplotlib.pyplot as plt

t = np.linspace(0,6,600)
x = np.exp(-t)

delay = np.where(t>=2,np.exp(-(t-2)),0)

plt.figure(figsize=(8,4))

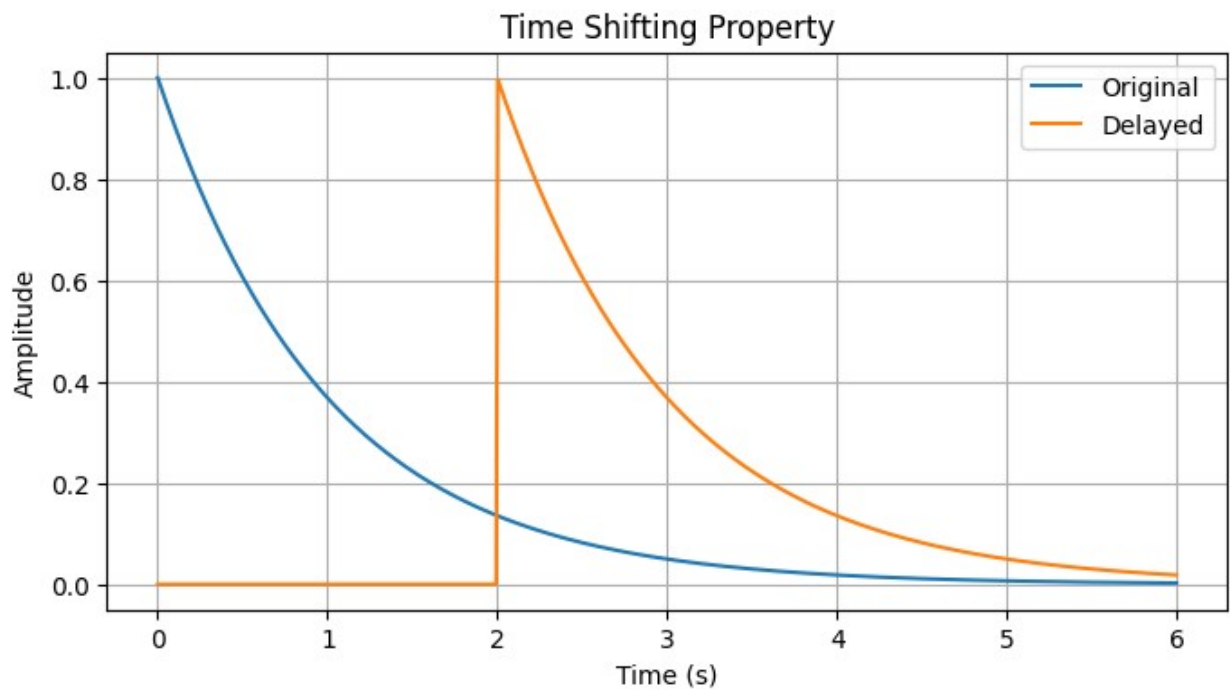
plt.plot(t,x,label='Original')
plt.plot(t,delay,label='Delayed')
```

```
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.title("Time Shifting Property")

plt.legend()

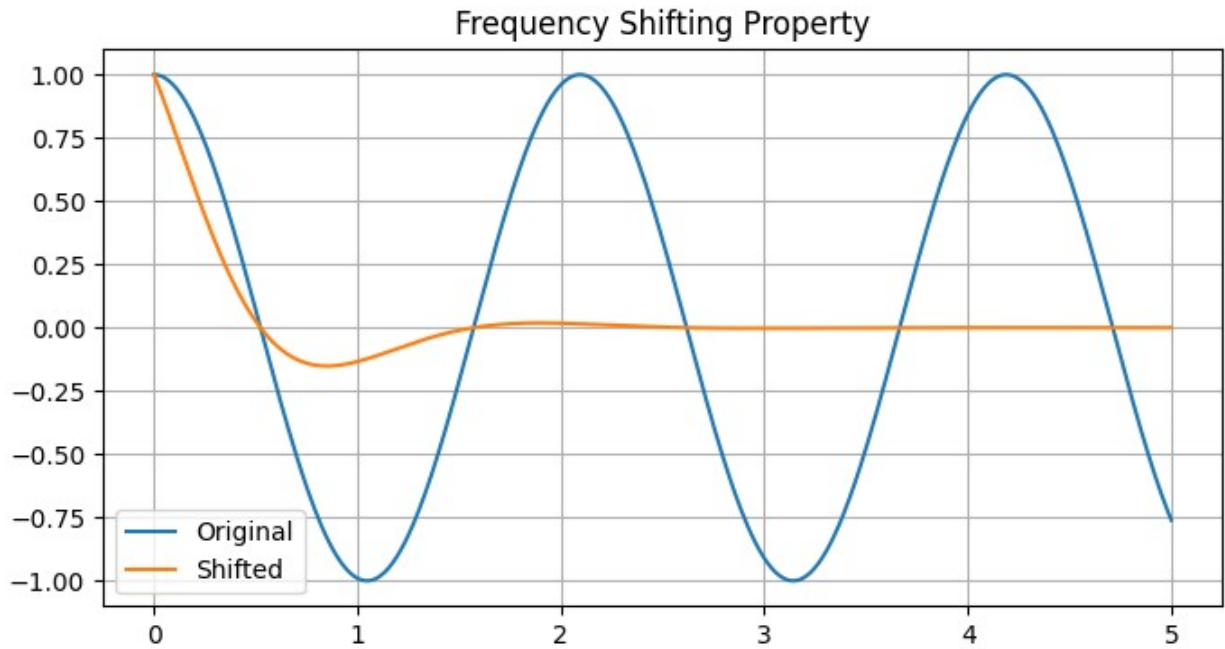
plt.grid(True)

plt.show()
```



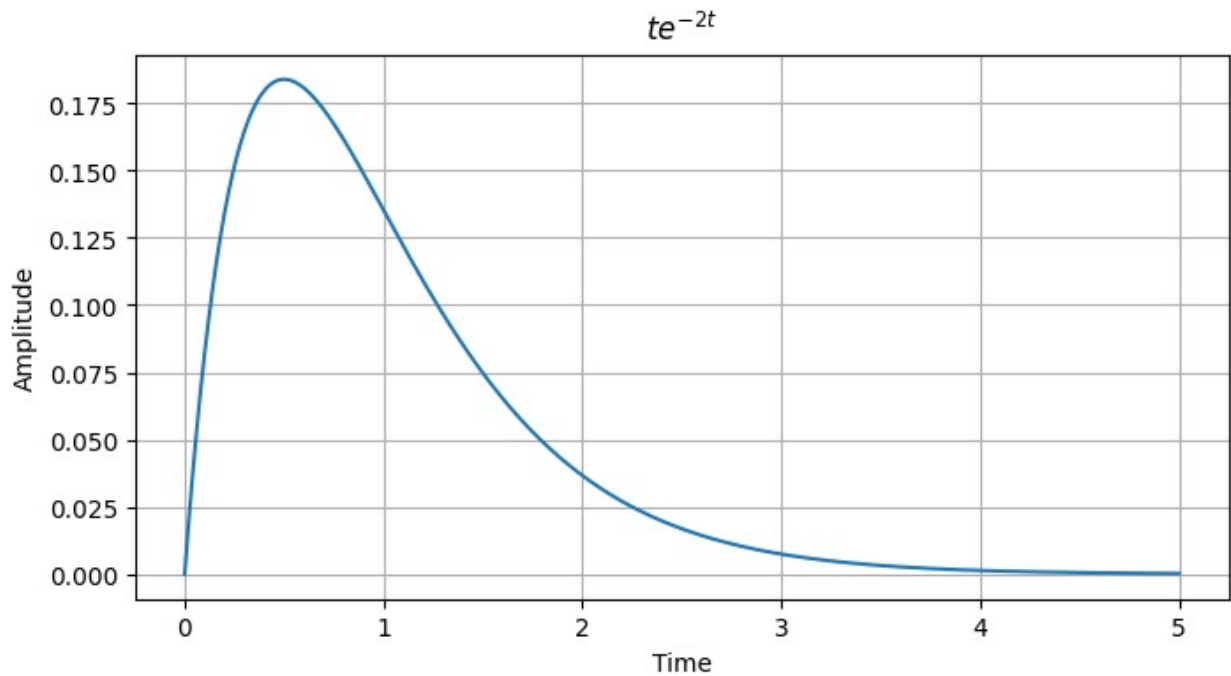
Problem 8

```
import numpy as np
import matplotlib.pyplot as plt
t=np.linspace(0,5,500)
original=np.cos(3*t)
shifted=np.exp(-2*t)*np.cos(3*t)
plt.figure(figsize=(8,4))
plt.plot(t,original,label="Original")
plt.plot(t,shifted,label="Shifted")
plt.grid(True)
plt.legend()
plt.title("Frequency Shifting Property")
plt.show()
```



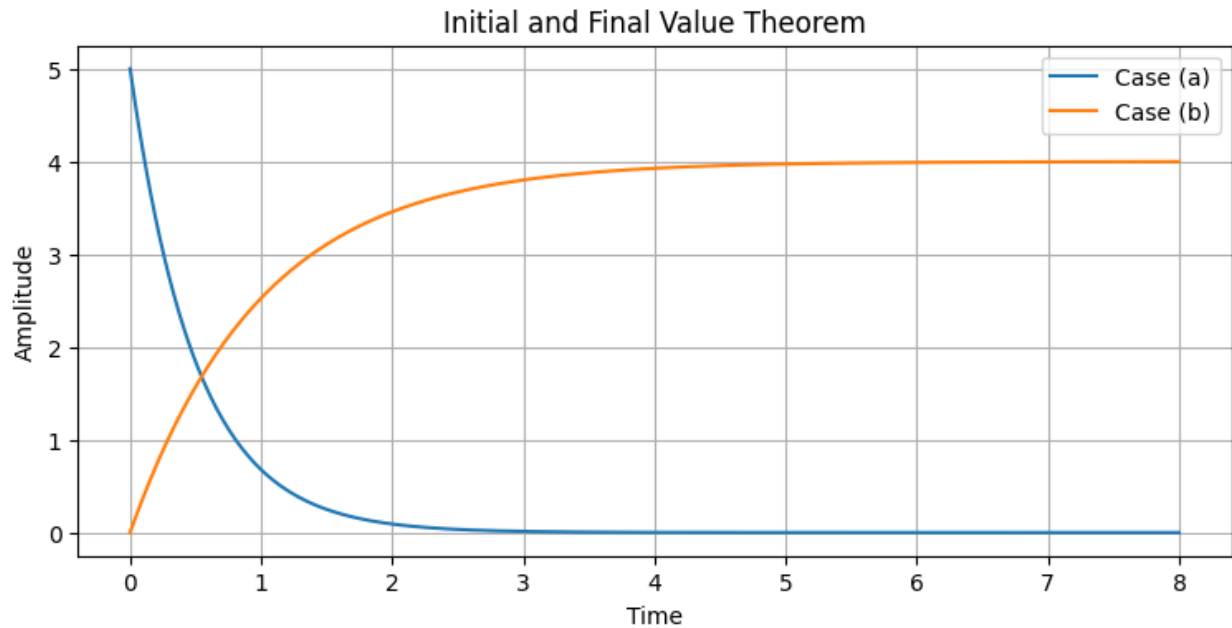
Problem 9

```
import numpy as np
import matplotlib.pyplot as plt
t=np.linspace(0,5,500)
x=t*np.exp(-2*t)
plt.figure(figsize=(8,4))
plt.plot(t,x)
plt.title(r"$te^{-2t}$")
plt.xlabel("Time")
plt.ylabel("Amplitude")
plt.grid(True)
plt.show()
```



Problem 10

```
import numpy as np
import matplotlib.pyplot as plt
t=np.linspace(0,8,500)
x1=5*np.exp(-2*t)
x2=4*(1-np.exp(-t))
plt.figure(figsize=(9,4))
plt.plot(t,x1,label="Case (a)")
plt.plot(t,x2,label="Case (b)")
plt.xlabel("Time")
plt.ylabel("Amplitude")
plt.title("Initial and Final Value Theorem")
plt.grid(True)
plt.legend()
plt.show()
```



Section C – Programming Problems

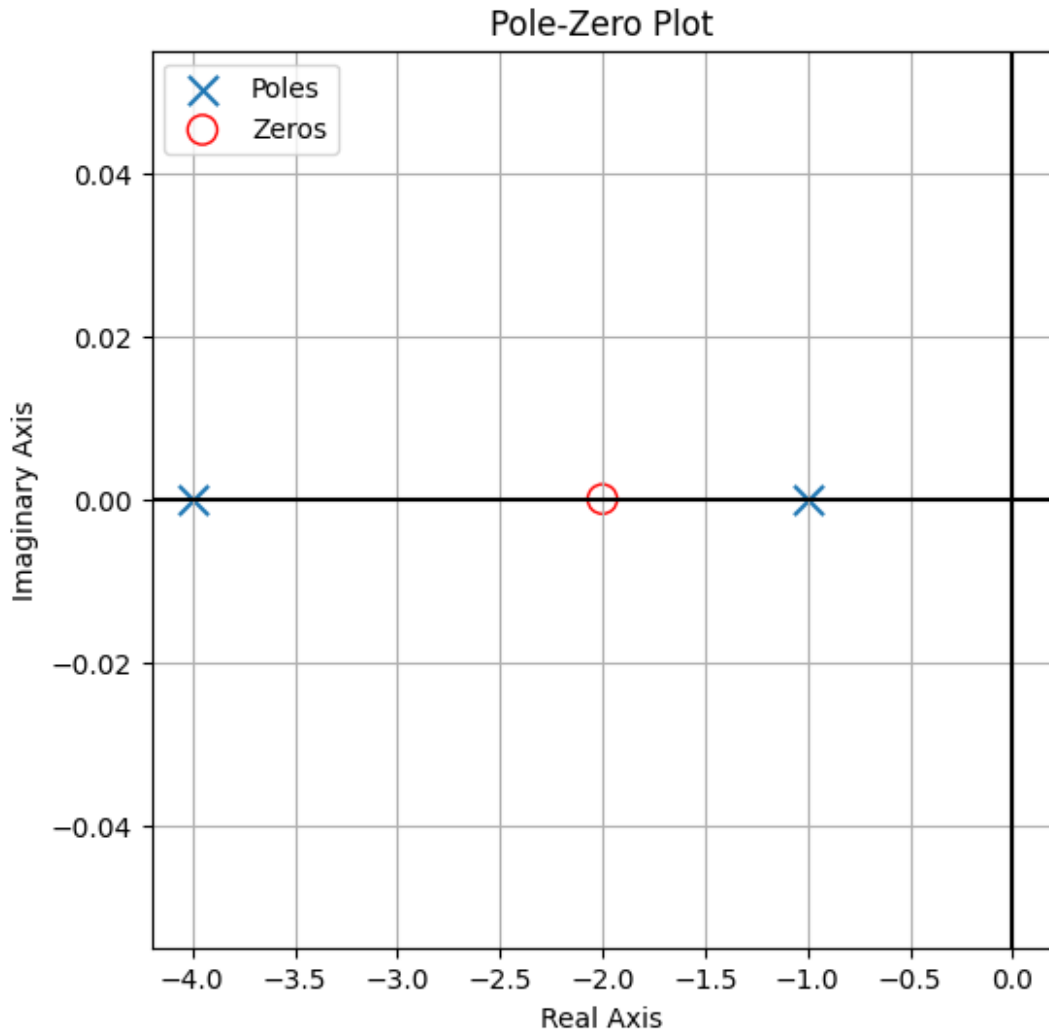
Problem 12

Pole: $s = -2$

Zero: $s = -1, -4$

The system is stable.

```
import matplotlib.pyplot as plt
poles = [-1, -4]
zeros = [-2]
plt.figure(figsize=(6,6))
plt.scatter(poles, [0,0], marker='x', s=120, label='Poles')
plt.scatter(zeros, [0], marker='o', s=120, facecolors='none',
edgecolors='red', label='Zeros')
plt.axhline(0,color='black')
plt.axvline(0,color='black')
plt.xlabel("Real Axis")
plt.ylabel("Imaginary Axis")
plt.title("Pole-Zero Plot")
plt.grid(True)
plt.legend()
plt.show()
```



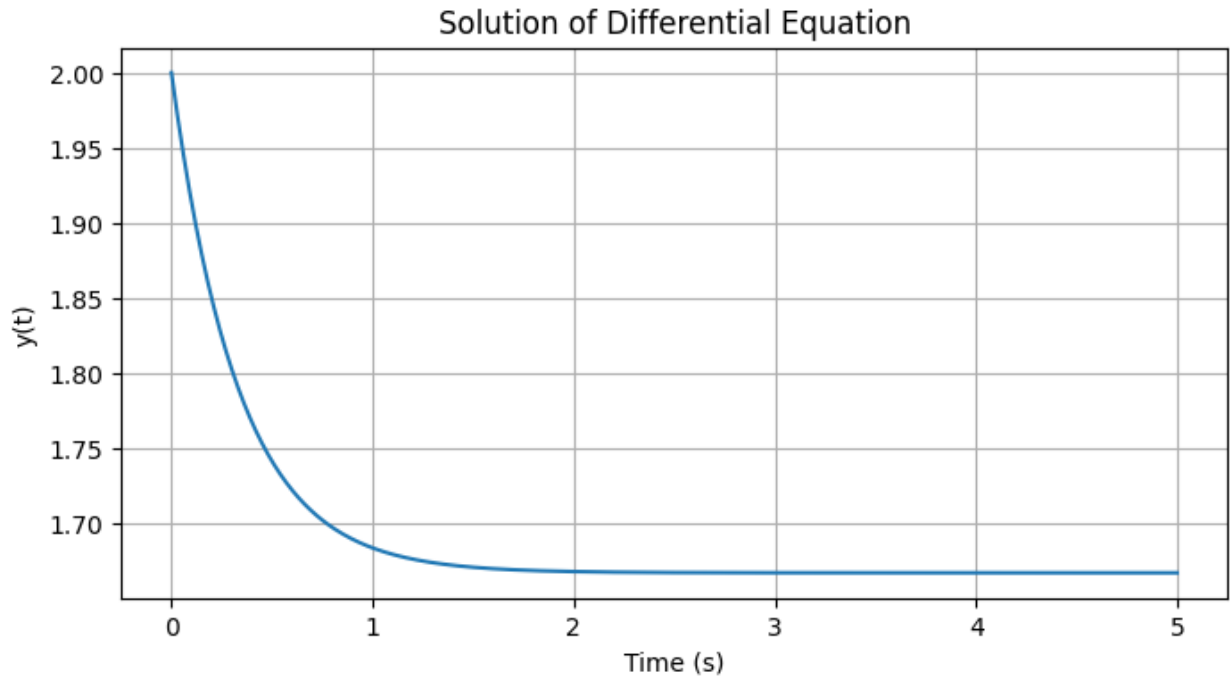
Problem 13

$$Y(s) = \frac{5}{s(s+3)} + \frac{2}{s+3}$$

Inverse Laplace:

$$y(t) = \frac{5}{3} + \frac{1}{3} e^{-3t}$$

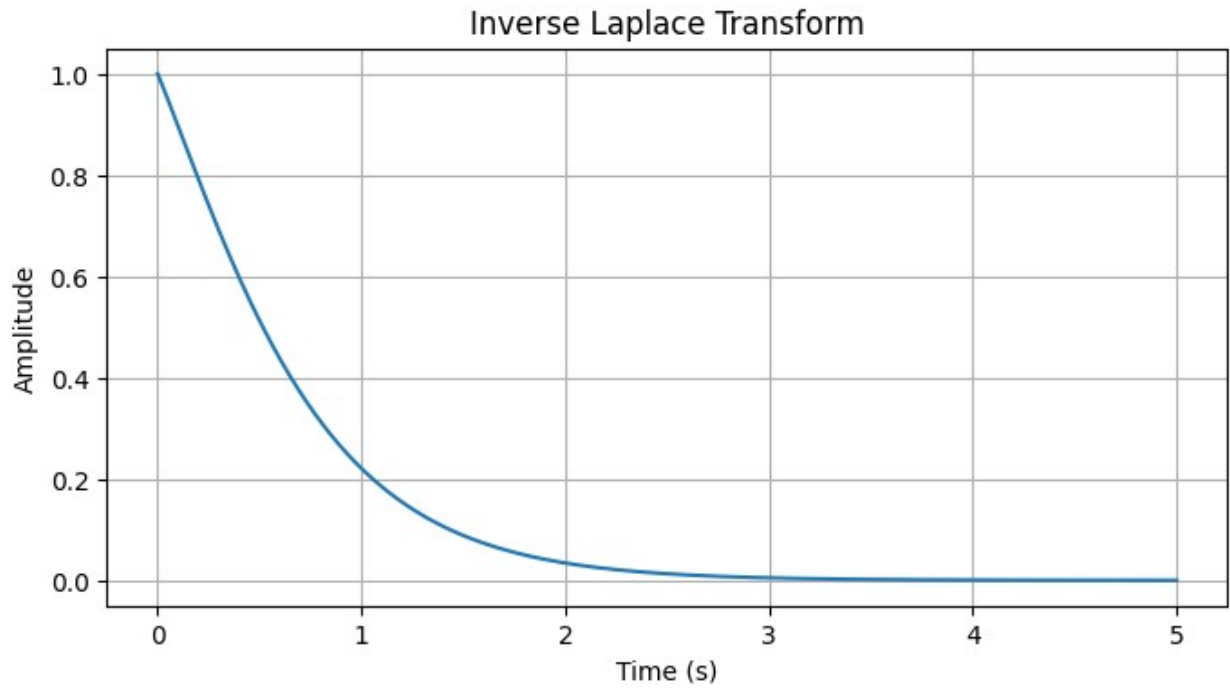
```
import numpy as np
import matplotlib.pyplot as plt
t=np.linspace(0,5,500)
y=5/3+(1/3)*np.exp(-3*t)
plt.figure(figsize=(8,4))
plt.plot(t,y)
plt.title("Solution of Differential Equation")
plt.xlabel("Time (s)")
plt.ylabel("y(t)")
plt.grid(True)
plt.show()
```



Problem 14

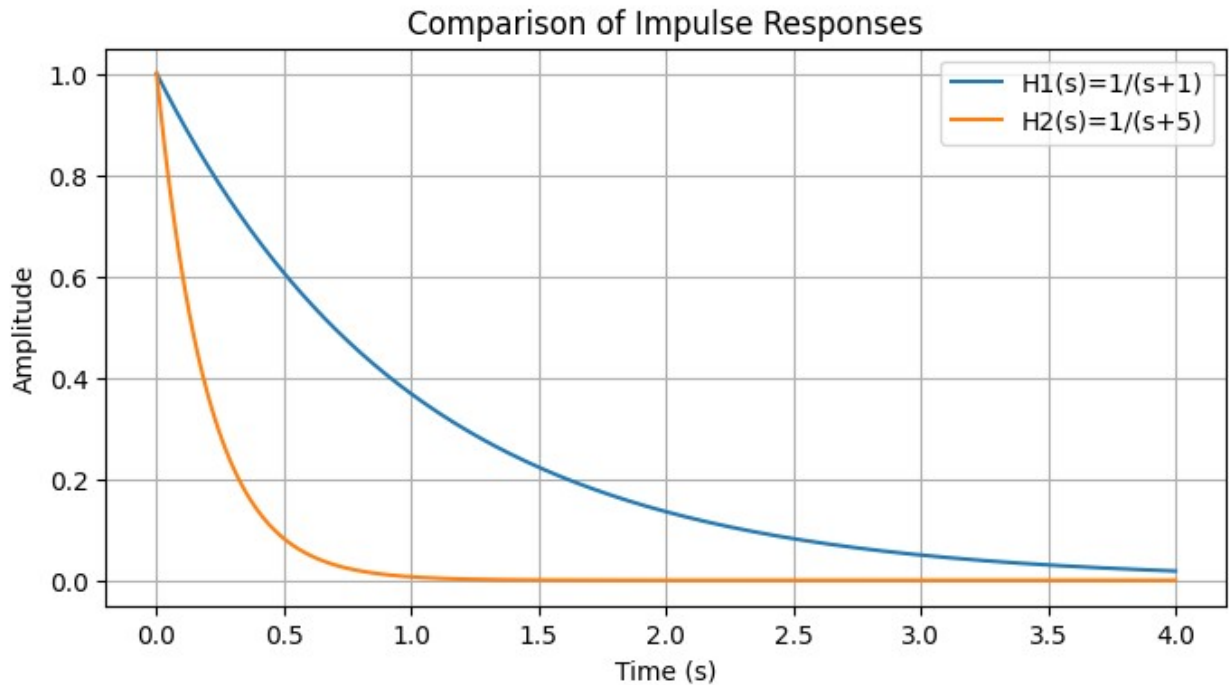
$$x(t) = 2e^{-2t} - e^{-3t}$$

```
import numpy as np
import matplotlib.pyplot as plt
t=np.linspace(0,5,500)
x=2*np.exp(-2*t)-np.exp(-3*t)
plt.figure(figsize=(8,4))
plt.plot(t,x)
plt.title("Inverse Laplace Transform")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid(True)
plt.show()
```



Problem 15

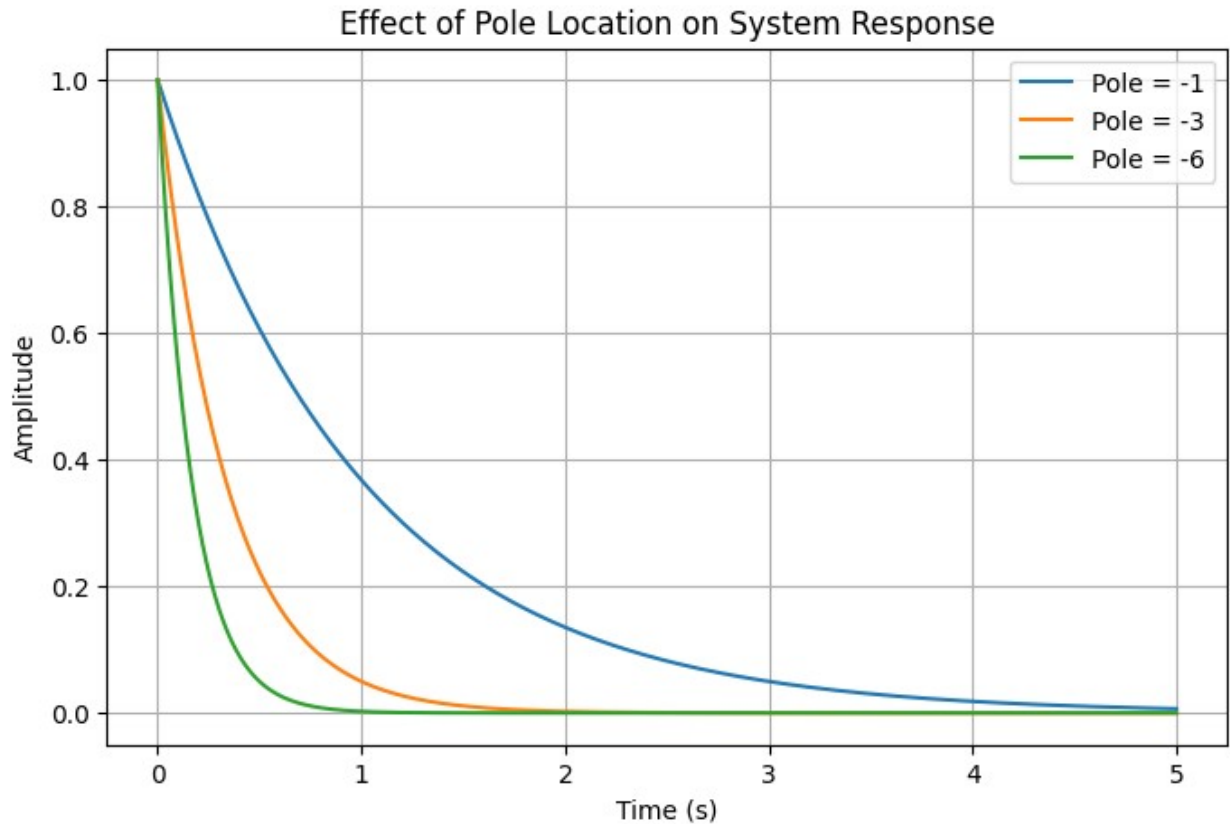
```
import numpy as np
import matplotlib.pyplot as plt
t=np.linspace(0,4,500)
h1=np.exp(-t)
h2=np.exp(-5*t)
plt.figure(figsize=(8,4))
plt.plot(t,h1,label="H1(s)=1/(s+1)")
plt.plot(t,h2,label="H2(s)=1/(s+5)")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.title("Comparison of Impulse Responses")
plt.grid(True)
plt.legend()
plt.show()
```



Challenge Problems

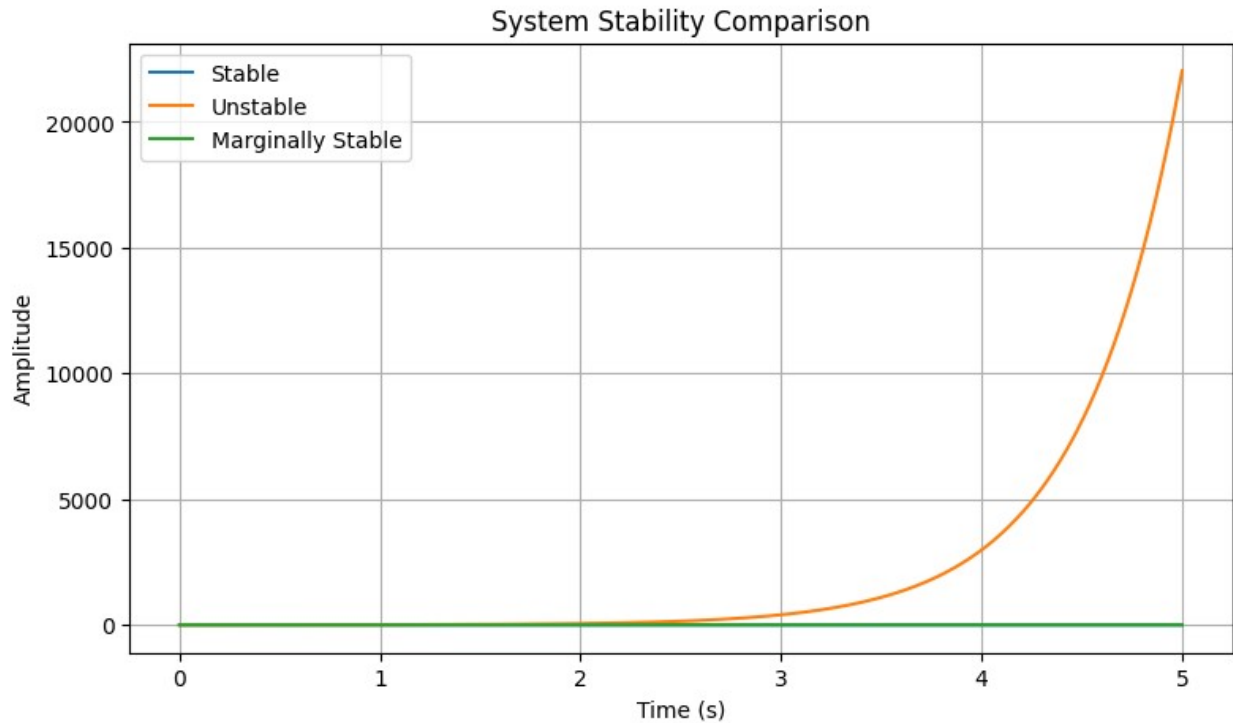
Challenge Problem 1

```
import numpy as np
import matplotlib.pyplot as plt
t = np.linspace(0,5,500)
responses = {
    "Pole = -1": np.exp(-t),
    "Pole = -3": np.exp(-3*t),
    "Pole = -6": np.exp(-6*t)
}
plt.figure(figsize=(8,5))
for label, y in responses.items():
    plt.plot(t, y, label=label)
plt.title("Effect of Pole Location on System Response")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid(True)
plt.legend()
plt.show()
```



Challenge Problem 2

```
import numpy as np
import matplotlib.pyplot as plt
t = np.linspace(0,5,500)
stable = np.exp(-2*t)
unstable = np.exp(2*t)
marginal = np.cos(2*t)
plt.figure(figsize=(9,5))
plt.plot(t, stable, label="Stable")
plt.plot(t, unstable, label="Unstable")
plt.plot(t, marginal, label="Marginally Stable")
plt.title("System Stability Comparison")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid(True)
plt.legend()
plt.show()
```



Challenge Problem 3

```
import matplotlib.pyplot as plt

# Example poles and zeros
poles = [-1+1j, -1-1j, -4]
zeros = [-2]

plt.figure(figsize=(6,6))

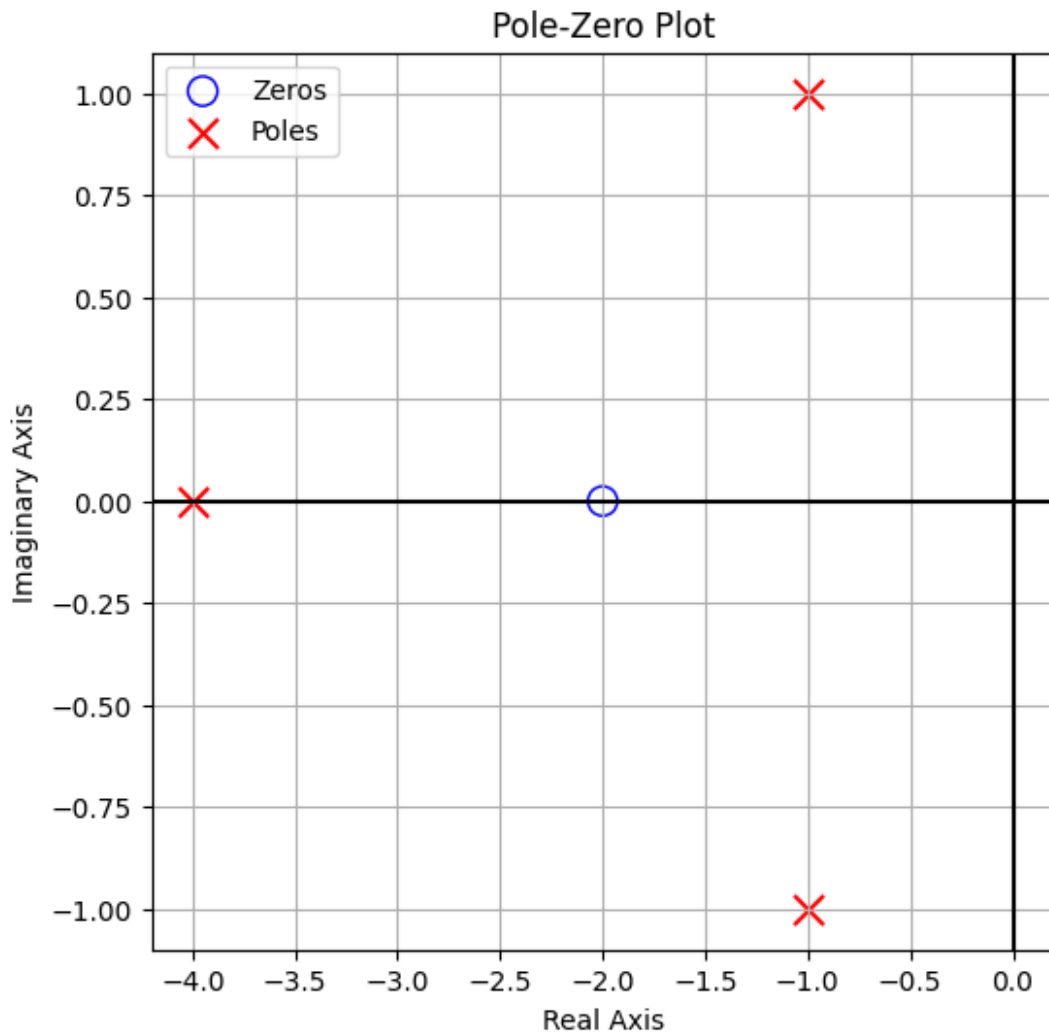
plt.scatter([z.real for z in zeros],
            [z.imag for z in zeros],
            marker='o',
            s=120,
            facecolors='none',
            edgecolors='blue',
            label='Zeros')

plt.scatter([p.real for p in poles],
            [p.imag for p in poles],
            marker='x',
            s=120,
            color='red',
            label='Poles')

plt.axhline(0,color='black')
plt.axvline(0,color='black')
```

```
plt.xlabel("Real Axis")
plt.ylabel("Imaginary Axis")
plt.title("Pole-Zero Plot")

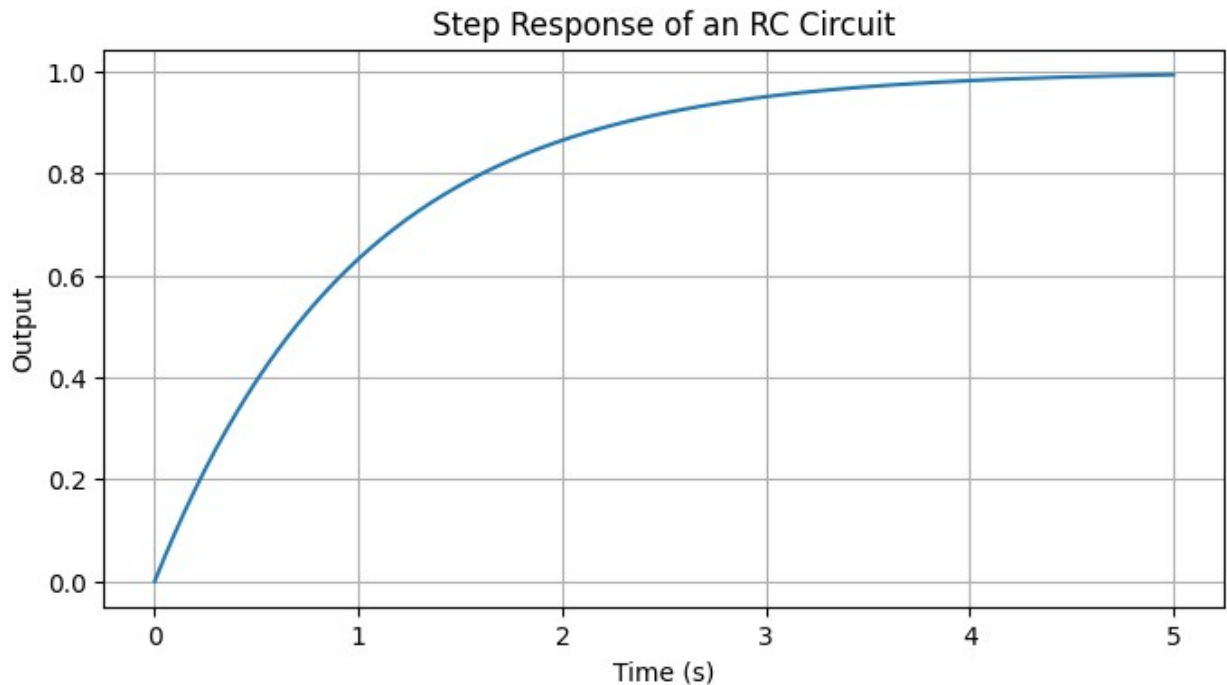
plt.grid(True)
plt.legend()
plt.show()
```



Challenge Problem 4

```
import numpy as np
import matplotlib.pyplot as plt
t = np.linspace(0,5,500)
y = 1 - np.exp(-t)
plt.figure(figsize=(8,4))
plt.plot(t,y)
plt.title("Step Response of an RC Circuit")
plt.xlabel("Time (s)")
```

```
plt.ylabel("Output")
plt.grid(True)
plt.show()
```



Challenge Problem 5

```
import numpy as np
import matplotlib.pyplot as plt
t = np.linspace(0,5,500)
analytical = np.exp(-2*t)
dt = t[1]-t[0]
numerical = np.zeros_like(t)
numerical[0] = 1
for i in range(len(t)-1):
    numerical[i+1] = numerical[i] + dt*(-2*numerical[i])
plt.figure(figsize=(8,4))
plt.plot(t, analytical, label="Analytical")
plt.plot(t, numerical, '--', label="Numerical")
plt.title("Analytical vs Numerical Solution")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.legend()
plt.grid(True)
plt.show()
```

